



DELHI PUBLIC SCHOOL NEWTOWN
SESSION: 2023-24
MONDAY TEST

CLASS : X
SUBJECT : CHEMISTRY

FULL MARKS : 40
DATE: 08.05.2023

General Instructions:

- The paper consists of three printed pages.
- Answers should be to the point.
- Question numbers should be copied carefully while answering the questions.
- Underline the keywords

Question: 1

Choose the correct answer from the options given below:

[5]

- (i) The hydroxide which is soluble in excess of ammonium hydroxide is :
A. Ferrous hydroxide
B. Magnesium hydroxide
C. Zinc hydroxide
D. Lead hydroxide
- (ii) The nature of bonding in phosphorus trichloride :
A. Ionic
B. Molecular
C. Coordinate
D. Covalent
- (iii) The energy required to remove an electron from a neutral isolated gaseous atom and convert it into a positively charged ion is called :
A. Electron affinity
B. Catenation
C. Electronegativity
D. Ionisation energy
- (iv) The noble gas with three shells:
A. Helium
B. Neon
C. Argon
D. Krypton
- (v) State the empirical formula of a compound whose molecular formula is $C_8H_6O_2$.
A. C_2H_3O
B. C_4H_3O
C. C_2HO
D. $C_4H_3O_2$

Question: 2

Name the following: [5]

- (i) A covalent compound that has one lone pair of electrons.
- (ii) A trivalent metal which reacts with hot and concentrated caustic soda to liberate hydrogen gas.
- (iii) A cation that does not form a precipitate with ammonium hydroxide but forms one with sodium hydroxide.
- (iv) An element in period two whose electron affinity is zero.
- (v) The type of bond formed when the combining atoms have zero electronegativity difference.

Question: 3

Give reasons for the following: [5]

- (i) Alkali metals are strong reducing agents.
- (ii) Noble gases have very high ionisation potential.
- (iii) Covalent compounds exist as gases, liquids or soft solids.
- (iv) Ionic compounds have high melting point.
- (v) Lead hydroxide dissolves in excess sodium hydroxide solution.

Question: 4

Solve the following numericals: [5]

- (i) A hydrocarbon on analysis was found to contain 2.2% hydrogen, 26.6% carbon and 71.2% oxygen. If the relative molecular mass of the compound is 90, determine its molecular formula. [Atomic mass: C = 12, H = 1, O = 16]
- (ii) Calculate the percentage of water of crystallisation in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$. [Cu=64, S=32, O=16, H=1]

Question: 5

Given below is a section of the Periodic Table. H does not represent hydrogen. Some elements are given in their own symbol and position in the periodic table while others are shown with a letter [5]

Group number	IA 1	IIA 2	IIIA 13	IVA 14	VA 15	VIA 16	VIIA 17	0 18
Period 2	Li		D			O	I	Ne
Period 3	A	Mg	E	Si		H	M	
Period 4	B	T	R		Q	U		Y

Answer the questions that follow:

- (i) Identify the element from period 3 with least atomic size.
- (ii) The element which is the most electronegative.
- (iii) The number of valence electrons present in Q.
- (iv) Write the formula of the compound formed between A and H.
- (v) What type of bond is formed between T and I?

Question: 6

(a) Draw the electron dot and cross structure to represent the following: [3]

- (i) Magnesium oxide formation
- (ii) Hydronium ion formation
- (iii) Ammonia formation

- (b) An element has 13 protons and 3 shells. [2]
- (i) How many valence electrons does this element have?
 - (ii) To which group does it belong?

Question: 7

- (a) Arrange the following as per the instructions given in the brackets () [3]
- (i) Cl, N, O
(decreasing order of number of covalent bond)
 - (ii) Na, Li, K
(increasing order of metallic character)
 - (iii) B, Li, F, O, C
(increasing order of electronegativity)
- (b) Write a balanced chemical equation for each of the following: [2]
- (i) Ammonium hydroxide solution is added dropwise to aqueous lead nitrate solution.
 - (ii) Zinc oxide is treated with hot and concentrated potassium hydroxide solution.

Question: 8

- (a) Differentiate between the following on the basis of the instruction given in the bracket: [3]
- (i) Halogens and Alkaline Earth metals *(position in the periodic table)*
 - (ii) Period 5 and Period 6 *(number of shells)*
 - (iii) Ferrous sulphate and Ferric chloride solutions *(colour of precipitate formed on reaction with sodium hydroxide)*
- (b) What would you observe when: [2]
- (i) Ammonium hydroxide solution is added to aqueous copper sulphate solution in little and then in excess.
 - (ii) Sodium hydroxide solution is added to aqueous calcium nitrate solution in excess.